

High-Dimensional Data Analysis Project

Group 3H

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1 Introduction

The analysis of stock returns is important in finance because it helps explain the relationship between risk and return. This report analyses monthly returns of New York Stock Exchange (NYSE) companies from 2005 to 2019 using high-dimensional data techniques. The study separates stock returns into systematic risk driven by overall market movements and idiosyncratic risk specific to individual firms or industries.

The objective is to investigate whether latent factors correspond to observable market benchmarks such as Market Return, SMB, and HML. Specifically, the report examines whether the first principal component represents the market, whether the Fama-French three-factor model explains common variation, which industries are most exposed to common factors, and whether any industry provides defensive or hedging characteristics.

2 Data and Preparation

Data preparation is the foundation of high-dimensional analysis. We utilize `SampleH.csv` (stock returns) and `MarketFactors.csv` (benchmarks).

Table 1: Sample of Dataset

Stock ID	Industry	Return	Market Return	SMB	HML
F85464	Wholesale Trade	-0.028	-2.76	-1.66	2.06
H89060	Finance, Insurance and Real Estate	-0.069	-2.76	-1.66	2.06
H87476	Finance, Insurance and Real Estate	-0.040	-2.76	-1.66	2.06
E79879	Transportation and Public Utilities	-0.322	-2.76	-1.66	2.06
H10777	Finance, Insurance and Real Estate	-0.042	-2.76	-1.66	2.06
B89901	Mining	0.154	-2.76	-1.66	2.06

The final dataset contains 14,940 observations across 9 variables with no missing values. The data cover 83 NYSE stocks observed monthly over 180 months from January 2005 to December 2019. Industry classifications include Mining (B), Construction (C), Manufacturing (D), Transportation and Public Utilities (E), Wholesale Trade (F), Retail Trade (G), Finance, Insurance and Real Estate (H), and Services (I).

Variables include stock returns together with three benchmark market factors: Market Return, SMB (Small Minus Big), and HML (High Minus Low). SMB represents the size effect between small and large firms, while HML captures the value effect based on book-to-market ratios.

3 Preliminary Analysis (EDA)

Table 2: Group statistics by the ‘industries’

Industry	Average Monthly Return	Volatility	Number of Stocks
Manufacturing	1.43%	12.87%	9
Wholesale Trade	1.23%	11.36%	6
Services	1.17%	16.48%	9
Mining	1.14%	15.95%	4
Transportation and Public Utilities	1.03%	12.25%	12
Retail Trade	0.96%	13.57%	10
Construction	0.80%	12.58%	1
Finance, Insurance and Real Estate	0.77%	7.29%	32

Table 2 summarises the performance and risk of eight industry sectors based on their monthly returns and volatility:

- **Top Performers:** Manufacturing leads with the highest average monthly return at 1.43%, followed closely by Wholesale Trade at 1.23%.
- **Risk and Volatility:** Services is the most volatile sector (16.48%), while Finance, Insurance and Real Estate is the least volatile (7.29%).
- **Sector Representation:** The Finance sector has the largest number of stocks in this dataset (32), whereas Construction is represented by only a single stock.

Industry Return Distributions: 2005 – 2019

Boxplots show median, quartiles, and volatility changes across industries

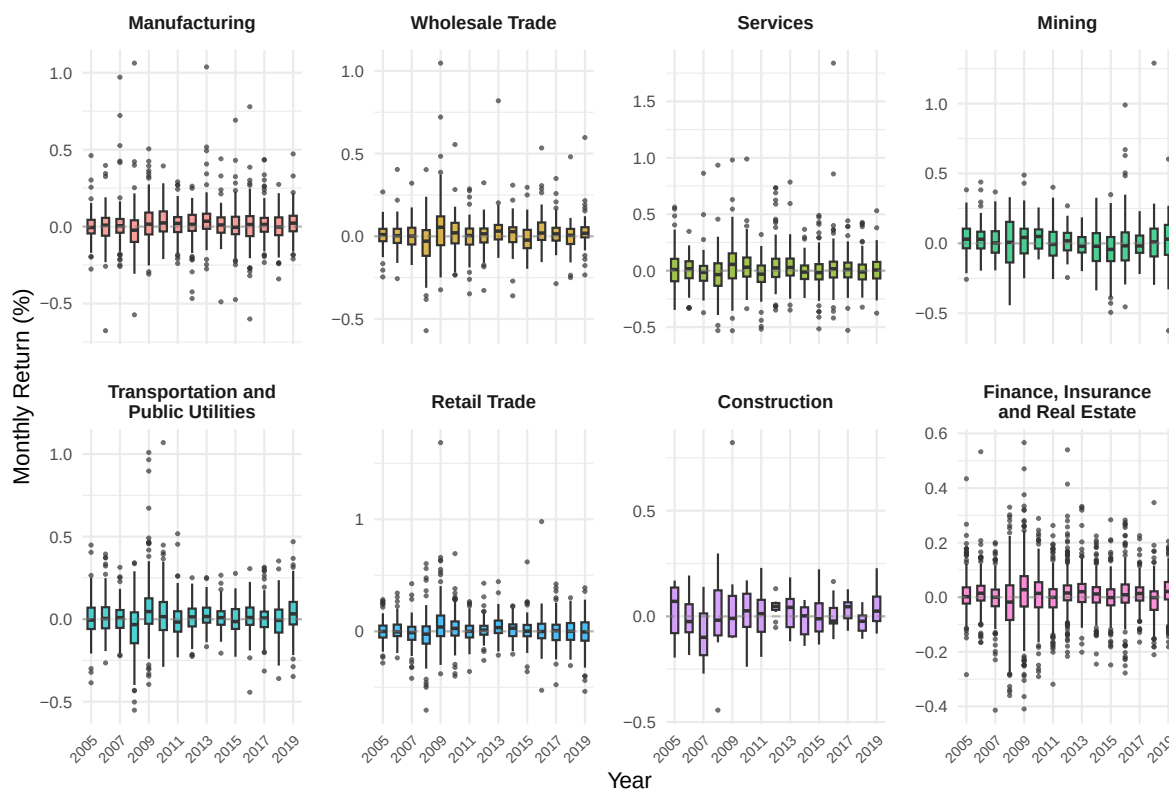


Figure 1: Distribution of monthly returns across industries from 2005 to 2019

As shown from the Figure 1, we can see the industry return distributions from 2005 to 2019 reveal several key trends in market stability and sector performance:

- **Volatility Cluster from 2008 to 2009:** During this time, all industries saw a big rise in volatility. We can see this in the taller boxes (wider interquartile ranges) and the extreme outliers, especially in Wholesale Trade, Services, and Finance.
- **Performance by sector:** Manufacturing and wholesale trade were still the best performers, with high average returns. However, manufacturing had more positive outliers over the decade.
- **Stability vs. Risk:** Finance, Insurance, and Real Estate had the most stable distributions (lowest volatility), while Mining and Services had much bigger swings from year to year.

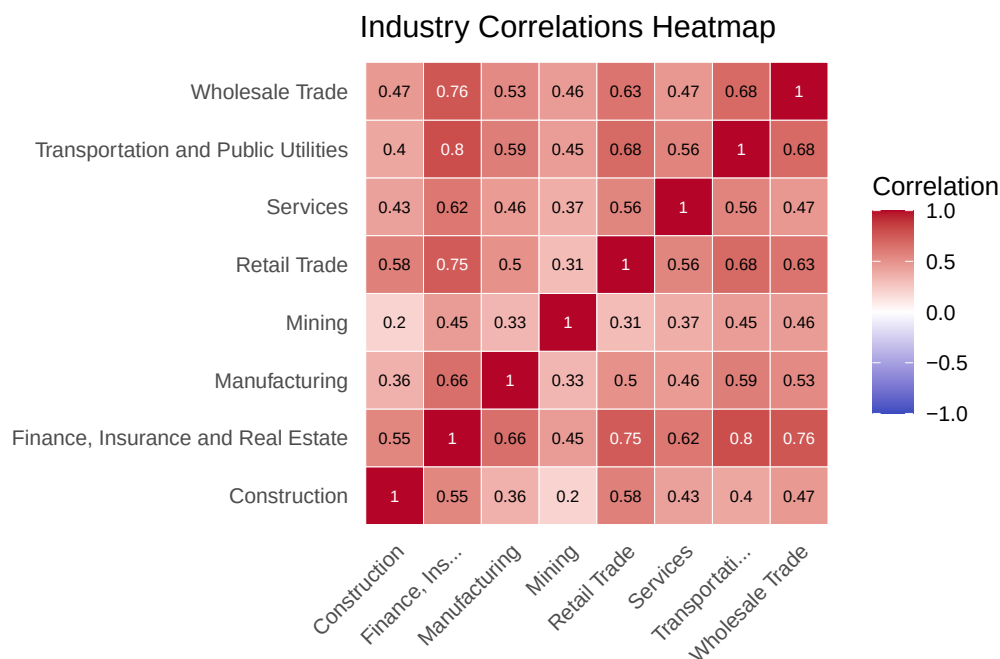


Figure 2: Industry Correlations Heatmap

Figure 2 shows preliminary correlation analysis indicates significant systemic integration between industries. For instance, the correlation coefficient between sectors Transportation and Public Utilities (E) and Finance, Insurance and Real Estate (H) is 0.80, highlighting a shared sensitivity to market conditions. Given this high dimensionality and strong inter-variable relationship, PCA is an appropriate methodology for isolating the primary component that parallel with broad market movements.

4 PCA Preparation

The selection of variables for PCA is limited to the 83 monthly stock return columns, as including the date or benchmark factors would bias the mathematical extraction of common variation. Since these stocks belong to diverse sectors with varying levels of price volatility, standardisation is a must. Without scaling, stocks with the highest return variance would disproportionately influence the principal components, regardless of their actual contribution to market-wide trends. By centering and scaling the data, we ensure each company contributes equally to the analysis. The PCA is then implemented on this standardised matrix to transform the correlated returns into a set of linearly uncorrelated principal components.

5 PCA and Latent Factors Analysis

5.1 Q1 Does the largest principal component coincide with market returns?

To determine the number of components representing meaningful signals, we generate a Scree Plot. In high-dimensional financial data, we typically observe a sharp elbow after the first component, which indicates that one dominant latent factor is responsible for the majority of the co-movement across all companies.

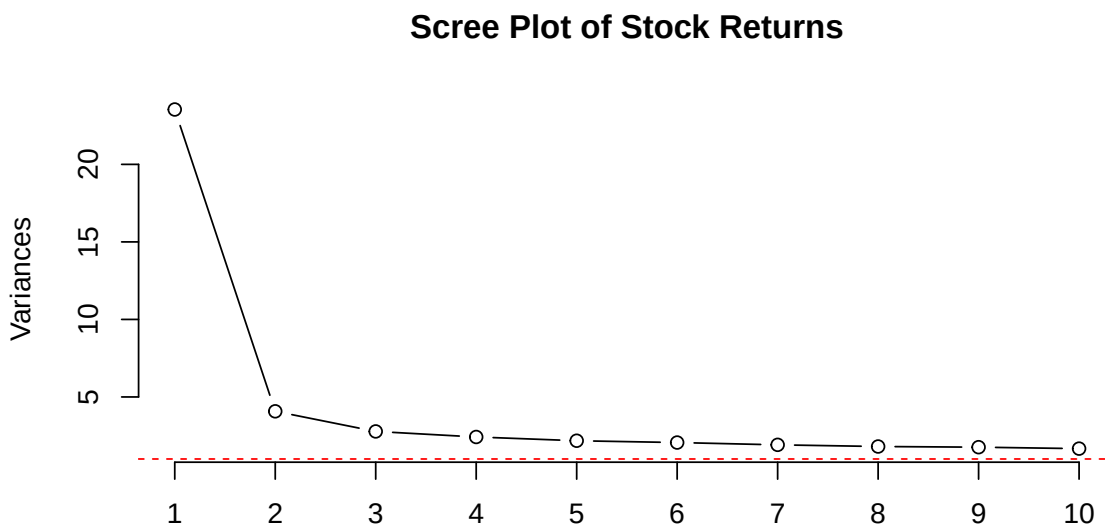


Figure 3: Scree Plot of Stock Returns

	PC1	PC2	PC3	PC4	PC5
Standard deviation	4.851678	2.018468	1.666329	1.555238	1.475485
Proportion of Variance	0.283600	0.049090	0.033450	0.029140	0.026230
Cumulative Proportion	0.283600	0.332690	0.366140	0.395280	0.421510

Scree Plot of Stock Returns

The Variance Explained by the first principal component (PC1) is a critical measure of systematic risk. If PC1 accounts for a large percentage (e.g., >30%) of the total variation in an 83-stock universe, it suggests a powerful common driver.

To test if this mathematical construct is in conformity with the actual economy, we compare the PC1 scores against the observable Market Return provided in the dataset.

Correlation between PC1 and Market Return: 0.9657

A correlation magnitude near or above 0.90 indicates compelling evidence that the first principal component (PC1) is a robust proxy for the broad market. While PCA is a purely mathematical algorithm with no prior knowledge of financial indices, the fact that its primary axis of variation aligns so closely with the broad market return confirms that the market is the single most significant latent force driving individual stock returns. Consequently, PC1 can be interpreted as the Systematic Risk Factor, representing macroeconomic events that affect the entire 83-stock universe together.

5.2 Q2 Do Market Return, SMB, and HML explain common variation in stock returns?

Table 3: Correlation between Principal Components and Market Factors

component	MarketReturn	SMB	HML
PC1	0.9657	0.4817	0.2852
PC2	0.0225	0.2410	0.4157
PC3	-0.0746	-0.0082	0.0495

To examine whether market factors explain common variation in stock returns, correlations were calculated between the principal components and Market Return, SMB, and HML. As shown in Table 3, PC1 has a very strong positive correlation with Market Return (0.9657), indicating that it captures broad market movement and represents the dominant source of common variation in stock returns. PC1 also shows moderate and weak relationships with SMB (0.4817) and HML (0.2852). In contrast, PC2 and PC3 have weak relationships with Market Return, although PC2 has a moderate relationship with HML (0.4157), suggesting some value-related effects.

Table 4: Correlation between latent factor scores and market factors

Latent factor	MarketReturn	SMB	HML
Factor1	0.9865	0.4136	0.1821
Factor2	0.0754	0.2399	0.4139
Factor3	-0.0027	0.1476	0.5170

Furthermore, Table 4 provides a consistency check using the latent factor model. Factor 1 also has a very strong relationship with Market Return (0.9865), confirming that the dominant hidden structure in stock returns is overall market movement. Factor 2 and Factor 3 show moderate relationships with HML, suggesting that SMB and HML explain only smaller secondary patterns.

5.3 Q3 Which industries move in the same direction as the common factors?

Table 5: Industry-level PC1 loadings

Industry	Mean PC1 Loading	SD of Loading	Min Loading	Max Loading	Number of Stocks
Finance, Insurance and Real Estate	0.118	0.050	0.005	0.199	32
Construction	0.117	NA	0.117	0.117	1
Wholesale Trade	0.107	0.021	0.086	0.144	6
Transportation and Public Utilities	0.107	0.033	0.049	0.158	12
Retail Trade	0.098	0.024	0.054	0.131	10
Manufacturing	0.082	0.042	0.031	0.144	9
Mining	0.079	0.036	0.040	0.110	4
Services	0.065	0.022	0.037	0.101	9

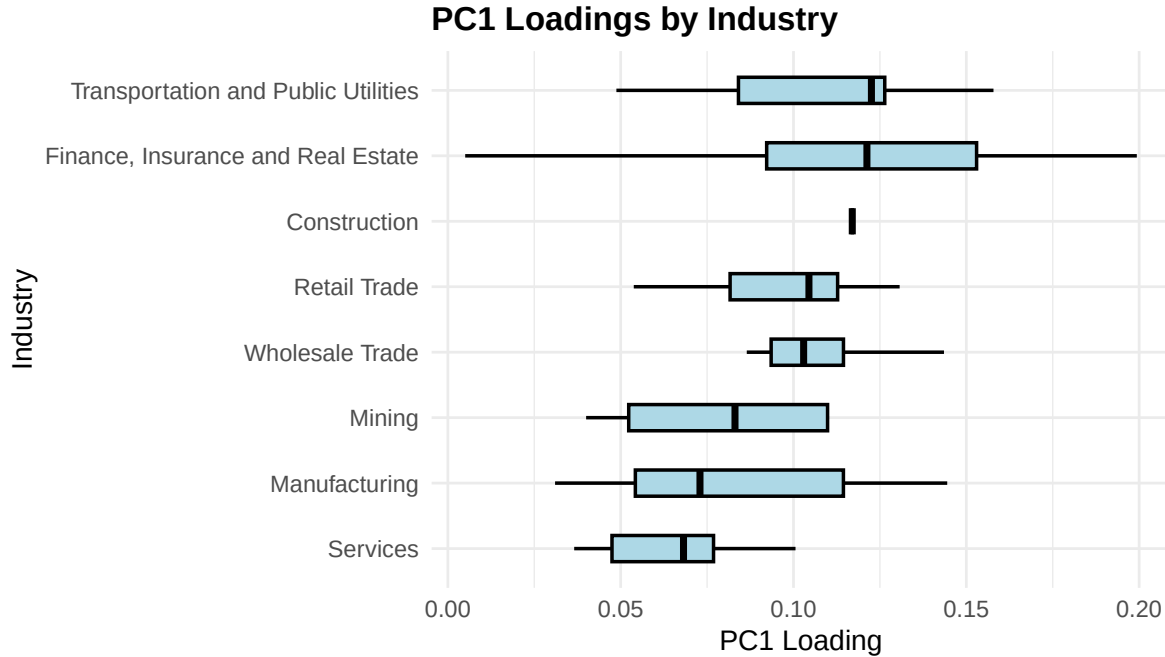


Figure 4: Distribution of PC1 loadings by industry

As shown in Table 5 and Figure 4, all industries have positive PC1 loadings, indicating that they move in the same direction as the dominant common factor. This confirms that the common factor affects the entire market rather than a specific group of industries.

However, the strength of this relationship differs across industries. Finance, Insurance and Real Estate have the highest average loading (0.118), followed closely by Construction (0.117) and Wholesale Trade (0.107), suggesting these industries are more sensitive to the common factor. In contrast, Services (0.065) and Mining (0.079) have lower average loadings, indicating weaker exposure.

More importantly, there is clear heterogeneity within industries. Finance, Insurance and Real Estate shows a relatively large spread ($SD = 0.050$ and a wide range from 0.005 to 0.199), indicating that stocks within this industry respond very differently to the common factor. Manufacturing and Mining also display noticeable variation, suggesting inconsistent behaviour across stocks. In contrast, Wholesale Trade and Retail Trade show smaller spreads, indicating more homogeneous responses.

The Construction industry appears as a single line in Figure 4 and has no variation in Table 5 because it contains only one stock. As a result, it is not meaningful to assess heterogeneity within this industry.

Overall, while all industries move in the same direction as the common factor, the key result is that the degree of heterogeneity varies substantially across industries, suggesting that industry classification alone does not fully explain stock behaviour.

5.4 Q4 Which industry has the highest systematic risk?

Table 6: Correlation between Principal Components and Market Factors

Industry	No of stocks	Mean PC1 correlation	Mean PC1 variance share
Finance, Insurance and Real Estate	32	0.571	0.382
Construction	1	0.568	0.322
Transportation and Public Utilities	12	0.520	0.293
Wholesale Trade	6	0.521	0.280
Retail Trade	10	0.473	0.236
Manufacturing	9	0.396	0.194
Mining	4	0.384	0.171
Services	9	0.317	0.111

To identify the industry with the highest systematic risk, we use PC1 as the main market-wide factor. This is reasonable because the earlier result showed that PC1 moves very closely with the actual Market Return.

Based on the results on Table 6, Finance, Insurance and Real Estate appears to have the highest systematic risk. It has the highest average PC1 variance share, at 0.382, and the highest average PC1 correlation, at 0.571, across 32 stocks. This suggests that returns in this industry are more strongly driven by the shared market movement than the other industries.

While Construction also has a relatively high PC1 variance share, at 0.322, but this result comes from only one stock. Because of that, it is risky to say as a whole industry conclusion. Finance, Insurance and Real Estate is the stronger answer because it has both high exposure to PC1 and a much larger number of stocks. This also makes sense in practice, because financial firms are usually sensitive to wider market conditions, investor confidence, credit and economic conditions.

5.5 Q5 Which industry is counter-cyclical for hedging purposes?

Table 7: Industry exposure to PC1 for hedging assessment

Industry	No of stocks	Mean PC1 correlation	Mean PC1 variance share	Signal
Services	9	0.317	0.111	Weak positive exposure
Mining	4	0.384	0.171	Weak positive exposure
Manufacturing	9	0.396	0.194	Weak positive exposure
Retail Trade	10	0.473	0.236	Positive exposure
Transportation and Public Utilities	12	0.520	0.293	Positive exposure
Wholesale Trade	6	0.521	0.280	Positive exposure
Construction	1	0.568	0.322	Positive exposure
Finance, Insurance and Real Estate	32	0.571	0.382	Positive exposure

No industry in this sample as shown Table 7 is truly counter-cyclical because all industries have positive average PC1 correlations. This means they still tend to move in the same direction as the market wide factor. However, Services is the most defensive option because it has the lowest average PC1 correlation (0.317) and the lowest average PC1 variance share (0.111). This suggests that Services is the least exposed to broad market movements, although it should not be treated as a true hedge that moves opposite to the market.

6 Conclusion

Based on the high-dimensional analysis of NYSE stock returns from 2005 to 2019, the study concludes that **a single dominant latent factor drives common market variation**.

Key Findings and Direct Answers

Q1: Market Coincidence: The first principal component (PC1) is a robust proxy for the market, showing a 0.9657 correlation with broad market returns.

Q2: Fama-French Model: While PC1 captures overall movement, SMB and HML factors explain only a smaller portion of variation, providing only partial support for the three-factor model.

Q3: Industry Direction: All analyzed industries move in the same direction as the common factor, confirmed by universally positive PC1 loadings.

Q4: Systematic Risk: Finance, Insurance and Real Estate carries the highest systematic risk, with the highest PC1 correlation (0.571) and variance share (0.382).

Q5: Hedging: No industry is truly counter-cyclical; however, Services is the most defensive option due to its minimal market exposure.

Investment Implications

Investors should recognize that diversification across these industries does not eliminate systematic risk, as all sectors remain positively correlated with the market. For hedging, the Services sector provides the best relative protection. Conversely, the Finance sector requires careful monitoring during macroeconomic shifts due to its high sensitivity to latent market forces.

Limitations and Assumptions

This analysis assumes that linear relationships and covariance structures are sufficient to capture common variation in stock returns. PCA and factor models also assume that historical return patterns remain informative over time. However, the analysis is limited to 83 NYSE stocks and three observable market factors, meaning additional latent influences may remain unexplained. Furthermore, industry representation is uneven, with some sectors containing relatively few stocks.

7 Reference

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7.1 AI Usage Statement

Generative AI was used to assist with selected parts of this assignment, mainly to improve coding efficiency, refine wording and grammar, and support the interpretation of several analytical concepts. AI suggestions were used to help produce cleaner and more efficient code, improve the clarity of explanations, and assist with interpreting PCA and factor models in relation to market movement. In addition, AI was used to support the analysis in understanding the relationship between principal components, Fama-French factors, industry movement, and the discussion of counter-cyclical industries. The main ideas, analytical decisions, coding implementation, interpretation, and final version of the assignment were still completed and reviewed by us.

- Link 1
- Link 2
- Link 3

8 Appendix

8.1 Data and Preparation Code

8.2 Preliminary Analysis Code

Table group statistics by the ‘industries’

Time-series boxplots

Heatmap plot

8.3 PCA Preparation Code

```
# 1. Select only stock return columns (numeric variables)
# We exclude 'date' and the Fama-French factors
pca_input <- stocks |> select(-Date)

# 2. Standardise and Run PCA
# 'scale. = TRUE' ensures data is standardised (mean=0, sd=1)
pca_res <- prcomp(pca_input, scale. = TRUE)
```

8.4 Q1 Code

Scree Plot

```
# 1. Visualize Eigenvalues
screeplot(pca_res, type = "lines", main = "Scree Plot of Stock Returns")
abline(h = 1, col="red", lty=2) # Kaiser Criterion threshold

# 2. Calculate Variance Explained
kable(summary(pca_res)$importance[, 1:5])
```

Comparison PC1 with Market Return

```
# 1. Extract PC1 scores
pc1_scores <- pca_res$x[, 1]

# 2. Correlate with Market Return
market_test_corr <- cor(pc1_scores, market_factors$MarketReturn)

cat("Correlation between PC1 and Market Return:", round(market_test_corr, 4))
```

8.5 Q2 Code

Table: Correlation between Principal Components and Market Factors

```
# Extract PCA scores
pc_scores <- as.data.frame(pca_res$x) |>
  mutate(Date = stocks$Date)
```

```

# Join PCA scores with market factors by Date
q2_data <- pc_scores |>
  inner_join(market_factors, by = "Date")

# Correlation table between first 3 PCs and market factors
q2_cor_table <- q2_data |>
  select(PC1, PC2, PC3, MarketReturn, SMB, HML) |>
  cor(use = "complete.obs") |>
  as.data.frame() |>
  rownames_to_column("component") |>
  filter(component %in% c("PC1", "PC2", "PC3")) |>
  select(component, MarketReturn, SMB, HML) |>
  mutate(across(where(is.numeric), ~ round(.x, 4)))

kable(q2_cor_table)

```

Table: factor correlation analysis

8.6 Q3 Code

Table: Industry-level PC1 loadings

```

pc1_loadings |>
  ggplot(aes(x = reorder(industry, pc1_loading, median), y = pc1_loading)) +
  geom_boxplot(fill = "lightblue", color = "black", width = 0.4, size = 0.7) +
  coord_flip() +
  labs(
    title = "PC1 Loadings by Industry",
    x = "Industry",
    y = "PC1 Loading"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(face = "bold"),
    axis.text.y = element_text(size = 10),
    axis.text.x = element_text(size = 10)
  )

```

Boxplot: Distribution of PC1 loadings by industry

```
pc1_loadings |>
  ggplot(aes(x = reorder(industry, pc1_loading, median), y = pc1_loading)) +
  geom_boxplot(fill = "lightblue", color = "black", width = 0.4, size = 0.7) +
  coord_flip() +
  labs(
    title = "PC1 Loadings by Industry",
    x = "Industry",
    y = "PC1 Loading"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(face = "bold"),
    axis.text.y = element_text(size = 10),
    axis.text.x = element_text(size = 10)
  )

```

8.7 Q4 Code

Industry risk based on PC1

```
pc1_sign <- ifelse(
  cor(pca_res$x[, 1], market_factors$MarketReturn, use = "complete.obs") < 0,
  -1,
  1
)

# Stock exposure to PC1
stock_pc1 <- tibble(
  stock_id = colnames(pca_input),
  industry_letter = substr(stock_id, 1, 1),
  industry_name = as.character(industry_lookup[industry_letter]),
  pc1_loading = pc1_sign * pca_res$rotation[, 1],
  pc1_corr = pc1_loading * pca_res$sdev[1],
  pc1_variance_share = pc1_corr^2
)

q4_industry_risk <- stock_pc1 |>
  group_by(industry_name) |>
  summarise(
    n_stocks = n(),
    mean_pc1_corr = mean(pc1_corr, na.rm = TRUE),

```



```

    mean_pc1_variance_share = mean(pc1_variance_share, na.rm = TRUE),
    .groups = "drop"
  ) |>
  arrange(desc(mean_pc1_variance_share)) |>
  mutate(across(-c(industry_name, n_stocks), ~ round(.x, 3))) |>
  rename(
    Industry = industry_name,
    `No of stocks` = n_stocks,
    `Mean PC1 correlation` = mean_pc1_corr,
    `Mean PC1 variance share` = mean_pc1_variance_share
  )

kable(q4_industry_risk)

```

8.8 Q5 Code

Industry exposure to PC1

```

q5_hedging <- stock_pc1|>
  group_by(industry_name)|>
  summarise(
    n_stocks = n(),
    mean_pc1_corr = mean(pc1_corr, na.rm = TRUE),
    mean_pc1_variance_share = mean(pc1_variance_share, na.rm = TRUE),
    .groups = "drop"
  ) |>
  arrange(mean_pc1_corr)|>
  mutate(
    signal = case_when(
      mean_pc1_corr < 0 ~ "Counter-cyclical",
      mean_pc1_corr < 0.40 ~ "Weak positive exposure",
      TRUE ~ "Positive exposure"
    )
  )|>
  mutate(across(-c(industry_name, n_stocks, signal), ~ round(.x, 3))) |>
  rename(
    Industry = industry_name,
    `No of stocks` = n_stocks,
    `Mean PC1 correlation` = mean_pc1_corr,
    `Mean PC1 variance share` = mean_pc1_variance_share,
    `Signal` = signal
  )

```

```
)  
kable(q5_hedging)
```